

# SatQuery AI - System Report

An agentic vision-language assistant for remote-sensing imagery. A controller reads the query and the input configuration, selects a specialist model, and returns an evidence-grounded answer with a visual overlay and an auditable execution trace.

## 1. System architecture

User query + image(s) > Controller (task + input routing) > Specialist model > Evidence + trace > Web UI

| LAYER                  | TECHNOLOGY                                 | STATUS      |
|------------------------|--------------------------------------------|-------------|
| Frontend               | React 19, Vite 8, Tailwind 4 (4 pages)     | Built       |
| Backend API            | FastAPI, single-image endpoint             | Built       |
| Branch 1: single image | Qwen2-VL-7B-Instruct + LoRA                | Not adapted |
| Branch 2: bi-temporal  | ResNet-18 U-Net + rule engine              | Measured    |
| Branch 3: optical-SAR  | Dual ResNet encoders + Qwen2.5-3B reasoner | Not trained |
| DeltaVLM adapter       | Interface only (NotImplementedError stubs) | Not built   |

Branches share one contract (*ChangeEvidence*) so the controller consumes a single object shape regardless of which specialist answered.

## 2. Branch 1 - single-image VQA and captioning

| ITEM              | SPECIFICATION                                                                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| Base model        | Qwen2-VL-7B-Instruct (vision-language)                                                                         |
| Adaptation        | LoRA / PEFT, rank 16, alpha 32, q/k/v/o projections                                                            |
| Quantisation      | 4-bit (8 GB VRAM) or fp16 (16 GB)                                                                              |
| Input             | GeoTIFF, TIFF, PNG, JPEG via rasterio / PIL                                                                    |
| Band handling     | RGB by colour interpretation; Sentinel B04/B03/B02 by name; otherwise first three bands, recorded in the trace |
| Task routing      | Keyword classifier -> captioning or VQA                                                                        |
| Outputs           | Answer or caption, model name, execution trace                                                                 |
| Evaluation        | RSVQA (exact match), VRSBench (BLEU, ROUGE-L, METEOR)                                                          |
| Measured accuracy | NOT YET MEASURED                                                                                               |

The pipeline, LoRA configuration and evaluation scripts are complete, but no adapter has been trained and no benchmark has been run. The problem statement requires remote-sensing adaptation explicitly, so this is mandatory scope.

## 3. Branch 2 - bi-temporal change analysis

The published CDVQA baseline reaches about 58% because it never uses semantic labels. This branch predicts what each changed pixel was and became, then applies the benchmark's rules deterministically.

| ITEM           | SPECIFICATION                                                  |
|----------------|----------------------------------------------------------------|
| Architecture   | U-Net, two decoders, ResNet-18 ImageNet encoder                |
| Parameters     | 21.6 M                                                         |
| Input / output | 6 channels (2 x RGB) at 512x512 -> two 7-class maps            |
| Training       | 40 epochs, AMP fp16, RTX 4050 6 GB, best epoch 22              |
| Answer engine  | Deterministic rules -> one of CDVQA's 19 answer tokens         |
| Inference      | 4 s model load, then about 200 ms per query                    |
| Interface      | One call in, evidence object out; registry descriptor included |
| Tests          | 312 passing                                                    |

## Measured results

| METRIC                      | VALUE  | NOTE                              |
|-----------------------------|--------|-----------------------------------|
| CDVQA Test average accuracy | 68.14% | 968 scenes, 39,686 questions      |
| CDVQA Test overall accuracy | 74.89% | official held-out split           |
| Published CDVQA baseline    | ~58%   | we are about 10 points above      |
| Majority-class floor        | 44.91% | what guessing alone scores        |
| Rule ceiling (oracle)       | 99.90% | rules on ground-truth maps        |
| Segmentation mIoU           | 42.08% | diagnostic, not the scored metric |

*The oracle figure proves the rules reproduce the benchmark's own answers almost exactly, so every remaining error is segmentation quality rather than reasoning. Near-duplicate screening bounds the true test figure at 67.7% or above.*

## 4. Branch 3 - optical-SAR paired analysis

Optical carries spectral and contextual detail; SAR penetrates cloud and works day or night. This branch learns a shared representation of the two so a query can draw on whichever modality actually carries the answer.

| ITEM              | SPECIFICATION                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------|
| Optical encoder   | ResNet-18/34, 13-channel first conv (all Sentinel-2 bands, not reduced to RGB)                        |
| SAR encoder       | ResNet-18/34, 2-channel (Sentinel-1 VV + VH backscatter)                                              |
| Embeddings        | 512-dim per modality, 256-dim projection heads                                                        |
| Alignment         | Symmetric InfoNCE (CLIP-style), temperature 0.07                                                      |
| Cloud handling    | Cloud-coverage-aware weighting: heavily clouded optical patches contribute less to the alignment loss |
| Fusion            | Concat + MLP (default) or cross-attention, 8 heads                                                    |
| Reasoning head    | Qwen2.5-3B-Instruct + LoRA, alpha 32, 64 visual tokens (8x8)                                          |
| Training config   | 50 epochs, batch 64, lr 3e-4, 5 warmup epochs                                                         |
| API               | FastAPI, /api/optical-sar endpoint                                                                    |
| Measured accuracy | NOT YET MEASURED                                                                                      |

*The architecture, losses, training loop and API are complete. Only a 3-epoch smoke run on synthetic data exists; retrieval scores there sit at chance, which is expected for synthetic input and is not a result. Training on real SEN12MS-CR data has not been run.*

## 5. Datasets

| DATASET        | ROLE                                                         | USED BY  | SCALE        |
|----------------|--------------------------------------------------------------|----------|--------------|
| SECOND         | Semantic change maps, training labels                        | Branch 2 | 4,662 pairs  |
| CDVQA          | Change VQA benchmark, 19 answers, 8 types                    | Branch 2 | 2,968 scenes |
| OSCD           | Real Sentinel-2 13-band pairs, GeoTIFF path                  | Branch 2 | 24 pairs     |
| SEN12MS-CR     | Paired Sentinel-1 SAR and Sentinel-2 optical, cloud-affected | Branch 3 | 13 + 2 bands |
| BigEarthNet-S2 | Domain adaptation corpus for LoRA                            | Branch 1 | subset       |
| RSVQA          | Single-image VQA evaluation                                  | Branch 1 | not yet run  |
| VRSBench       | Captioning evaluation                                        | Branch 1 | not yet run  |

## 6. Problem statement coverage

| MANDATORY REQUIREMENT                   | STATUS                     |
|-----------------------------------------|----------------------------|
| Single-image VQA                        | Built, not measured        |
| Second single-image task (captioning)   | Built, not measured        |
| Bi-temporal change analysis             | Built and measured         |
| Spatial change map                      | Built                      |
| Optical-SAR paired analysis             | Built, not trained         |
| Remote-sensing adaptation (fine-tuning) | NOT DONE                   |
| Agentic tool orchestration              | Partial                    |
| Confidence estimation                   | Built (branch 2, measured) |
| Visual evidence                         | Built (branch 2)           |
| Execution trace                         | Built (both branches)      |
| Downloadable report                     | Built (branch 2)           |
| Interactive web application             | Built                      |

## 7. What distinguishes this build

- Confidence is measured, never invented. Every number is an accuracy actually observed on validation for that question type and land-cover class. Where no measurement applies it reports 0.0 and states why.
- Out-of-distribution input is detected. A physics-based producer cross-checks the trained model; when they disagree the system withdraws its confidence rather than answering confidently.
- Refusals over fabrication. SAR carries no land-cover mapping, so class questions are refused. Ungeoreferenced input reports pixel counts, never an invented area.
- Full execution trace: every stage records the tool, parameters, observation, duration and why the step was taken.
- Visual evidence is built from the same pixels the rule counted, so the picture cannot contradict the text.

## 8. Known gaps

- Remote-sensing fine-tuning on branch 1 has not been run. Mandatory.
- Branch 3 optical-SAR is architecturally complete but has never been trained on real data, so it has no measured capability. Mandatory scope.
- Branch 1 has no measured accuracy on any benchmark.
- Branch 2 under-predicts rare classes: water at 0.21x true frequency, playgrounds never predicted. Declared in the tool descriptor and reflected in its confidence.
- Branch 2 does not transfer across sensors yet: trained on sub-metre aerial imagery, it detects nothing at Sentinel-2's 10 m and correctly reports zero confidence.
- The controller registry spanning both branches is not yet wired.

*All figures are measured on the stated splits. Nothing in this report is estimated or projected.*